FinSphere Calc

A PROJECT REPORT

Submitted by

Kishan Vipulbhai Vaishnav

(23030601049)

In partial fulfilment for the award of the degree of

Bachelor of Science in Information Technology (BSc IT)

at

DARSHAN INSTITUTE OF COMPUTER APPLICATION (DICA), RAJKOT



Darshan University, Rajkot

March – 2026

Darshan Institute of Computer Application, Rajkot

Rajkot - Morbi Hwy, Rajkot, Gujarat 363650

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **"FinSphere Calc"** has been carried out by "**Kishan Vipulbhai Vaishnav**" under my guidance in partial fulfilment for the degree of Bachelor of Science in Information Technology, 6th Semester of Darshan University, Rajkot during the academic year 2025-26.

Internal Guide
Dr. Sharon V. Mohtra
Assistant Professor,
DICA, Darshan University

Dean-DICA
Dr. Gopi Sanghani
Darshan University

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application, Rajkot
Rajkot - Morbi Hwy, Rajkot, Gujarat 363650

DECLARATION

We hereby declare that the Project report submitted along with the Project entitled **FinSphere Calc** submitted in partial fulfilment for the degree of **Bachelor of Science in Information Technology** to Darshan University, Rajkot, is a Bonafide record of original project work carried out by me at Darshan Institute of Computer Application under the supervision of **Dr. Sharon V. Mohtra** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student	Sign of Student
Kishan Vipul Bhai Vaishnav	_____

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to my project guide **Prof. Dr. Sharon V. Mohtra** and all the faculty members for helping me through my project by giving me the necessary suggestions and advices along with their valuable co-ordination in completing this work.

I also thank my parents, friends and all the members of the family for their precious support and encouragement which they had provided in completion of my work. In addition to that, I would also like to mention the college personals who gave me the permission to use and experience the valuable resources required for the project from the college premises.

Thus, in conclusion to the above said, I once again thank the faculties and members of **Darshan Institute of Computer Application** for their valuable support in completion of the project.

With Sincere Regards

Kishan Vipulbhai Vaishnav

ABSTRACT

FinSphere Calc

FinSphere Calc is a financial calculator application which aims to simplify complex financial calculations for users through a single digital platform. This system is designed to help users easily calculate loan instalments, investment returns, and other financial values without performing manual calculations. The application converts traditional financial calculations into a smart digital solution that provides quick and accurate results. This project includes multiple financial calculators such as Home Loan EMI Calculator, Personal Loan Calculator, Fixed Deposit (FD) Calculator, Recurring Deposit (RD) Calculator, and other financial tools which help users make better financial decisions. The system provides an easy-to-use interface where users can enter financial parameters such as loan amount, interest rate, and tenure, and instantly get results like monthly EMI, total interest payable, and total repayment amount. Additionally, the system provides graphical representation and comparative analysis, allowing users to compare different loan options and understand the distribution between principal and interest amounts. This helps users make informed financial decisions in a clear and visual manner.

Overall, the FinSphere Calc application acts as a complete digital financial calculator platform that improves accuracy, reduces manual effort, and provides users with a convenient way to perform financial planning and analysis.

LIST OF FIGURES

Figure No	Figure Description	Page No
3.1.1.1	Use case diagram	6
3.1.3.1	Activity diagram	9
3.1.4.1	Sequence Diagram	10
3.1.5.1	State diagram	11
3.2.2	ER Diagram	12
4.1.1	Login Screen	13
4.1.2	Signup Screen	14
4.13	Dashboard Screen	16
4.1.4	Calculator Screen	18
4.1.5	Pie Chart Screen	20
4.1.6	Date Difference Screen	21
4.1.7	Loan Comparison	22

LIST OF TABLES

Table No	Table Description	Page No
3.9.1	Admin	12
3.9.2	User	12
4.1.1	Screen Element of Login Screen	14
4.1.2	Screen Element of Signup Screen	15
4.1.3	Screen Element of Dashboard Screen	16
4.1.4	Screen Element of Calculator Screen	18
4.1.5	Screen Element of Pie Chart Screen	20
4.1.6	Screen Element of Date Difference Screen	21
4.1.7	Screen Element of Loan Comparison	22
4.2.1	Functional Testing	24
4.2.2	Non-Functional Testing	24

TABLE OF CONTENT

Chapter No	Chapter Title	Page No
	Acknowledgement	i
	Abstract	ii
	List of Figures	iii
	List of Tables	iv
	Table of Contents	vi
Chapter 1	**INTRODUCTION**	1
	1.1 Motivation	1
	1.2 Project Scope	1
	1.3 Project Stakeholders	1
	1.4 Project Scheduling	1
Chapter 2	**SYSTEM ANALYSIS**	2
	2.1 Introduction	2
	2.2 Requirement Analysis	2
	2.2.1 Functional Requirement	2
	2.2.2 Non-Functional Requirement	2
	2.2.3 Software & Hardware Requirements	2
	2.3 Design and Implementation Constraints	3
	2.4 Assumptions and Dependencies	3
Chapter 3	**SYSTEM DESIGN**	6
	3.1 UML Diagram	6
	3.1.1 Use case Diagram	6
	3.1.2 Use case Scenario	7
	3.1.3 Activity Diagram	9
	3.1.4 Sequence Diagram	10
	3.1.5 State Diagram	11
	3.2 Database Design	12
	3.2.1 Database Schema	12
	3.2.2 E-R Diagram	12
Chapter 4	**IMPLEMENTATION**	13
	4.1 Module Design and Specification	13
	4.2 Testing	24
Chapter 5	**CONCLUSION AND FUTURE ENHANCEMENT**	25
	5.1 Conclusion	25
	5.2 Future Enhancement	25
	APPENDICES	26
	REFERENCES	26

CHAPTER 1

INTRODUCTION

1.1 Motivation

A study of existing financial calculator systems often reveals several limitations such as the need to use multiple websites or applications for different financial calculations, lack of centralized functionality, limited comparison features, and poor visualization of financial data. Many traditional calculators provide only basic numerical results without offering detailed insights into financial structures such as the distribution between principal and interest. Additionally, some platforms have complex user interfaces that make it difficult for users to perform calculations quickly and efficiently.

These limitations create inconvenience for users who want to evaluate different financial options such as loans, deposits, or investments. Users often need to manually compare results from different platforms, which increases the chances of calculation errors and reduces decision-making efficiency.

The main objective of this document is to illustrate the requirements of the FinSphere Calc – Financial Calculator Application. This document describes the Software Requirements Specification (SRS) for the financial calculator system. It provides a detailed description of system functionalities, modules, and technical requirements needed for the development of the application.

1.2 Project Scope

The application allows users to calculate various financial parameters such as loan EMI, total interest payable, total repayment amount, fixed deposit returns, and recurring deposit maturity values. Instead of using multiple websites or performing manual calculations, users can access all financial tools within a single platform. Overall, this project aims to create a simple, efficient, and user-friendly financial decision-support system that can handle multiple financial calculations and assist users in planning their loans and investments effectively.

1.3 Project Stakeholders

The project stakeholders for this document are:

- Financial Analysts / Advisors (Admin)
- End Users (App Users)
- Software Project Managers
- Software Engineers
- Software Developers

1.4 Project Scheduling

Phase	Tasks	Duration
Week 1-2	Requirement Analysis and Feasibility Study for Financial Calculator Application	2 Weeks
Week 3	System Design (UML Diagrams, Wireframes, System Architecture)	1 Week
Week 4-5	Database Design and Setup using Firebase / Cloud Database	2 Weeks
Week 6-9	Frontend Development using Flutter (UI Design, Calculator Screens, Navigation)	4 Weeks
Week 10-12	Backend Integration (Firebase Authentication, Database Integration, Data Handling)	3 Weeks
Week 13-14	Implementation of Financial Calculators and Chart Visualization	2 Weeks
Week 15-16	System Integration and Testing	1 Week
Week 17	Deployment and Documentation	1 Week
Week 18	Final Review and Project Submission	1 Week

CHAPTER 2

SYSTEM ANALYSIS

2.1 Introduction

This project focuses on developing a digital financial calculator application that simplifies complex financial calculations and provides users with quick and accurate results. The system converts traditional manual financial calculations into a modern mobile-based application where users can easily perform different types of financial computations.

2.2 Requirement Analysis

2.2.1 Functional Requirement

Admin

- **Manage User Accounts:** Admin can view, update, or remove user accounts from the system.
- **Monitor System Usage**: Admin can monitor application usage and manage system activities**.**
- **Manage Financial Modules:** Admin can enable or disable different financial calculators available in the application.
- **View Reports:** Admin can generate reports related to user activity and calculation history**.**
- **Maintain System Data:** Admin ensures that the system database and stored financial records are maintained properly.

Registered User

- **Authentication:** User must be authenticated before accessing the application features.
- **Use Financial Calculators:** Users can perform different financial calculations such as EMI calculation, FD maturity value, and RD returns.
- **Input Financial Data:** Users can enter financial parameters such as loan amount, interest rate, and tenure to perform calculations.
- **View Calculation Results:** Users can view outputs like monthly EMI, total interest payable, and total repayment amount.
- **Compare Financial Options:** Users can compare different loan options or financial plans to analyse which option is more beneficial.
- **View Calculation History:** Users can view their previous financial calculations saved in the system.
- **Update Profile:** Users can update their personal information and account settings.

2.2.2 Non-Functional Requirement

Performance Requirements

- The system should provide financial calculation results instantly with minimal delay.
- The application should support multiple users performing calculations simultaneously without affecting performance.
- The response time for financial calculations should be less than 2 seconds.

Scalability

- The system should support the addition of new financial calculators and features in the future.
- The database should handle a large number of user accounts and calculation records efficiently.
- The system architecture should allow easy updates and expansion.

Security

- User authentication should be implemented using secure login mechanisms such as Firebase Authentication.
- User data and financial calculation history should be stored securely in the database.
- Only authorized users should be able to access or modify personal data.
- Secure data transmission protocols should be used to protect sensitive user information.

2.2.3 Software & Hardware Requirements

User	Particulars	Client System	Server System
User/ Admin (Mobile App)	Operating System	Android 6.0 and Higher / iOS 13 and Higher	Cloud Server
	Processor	Quad Core Processor (Minimum)	High Performance Cloud Processor
	Enteral Memory	100 MB Free Space (Minimum)	Scalable Cloud Storage
	RAM	2 GB (Minimum)	8 GB or Higher

2.3 Design and Implementation Constraints

All users of the application must keep their login credentials confidential to ensure the security of their personal data and financial calculation records. The system requires each user to have a unique user ID and password to access the application and perform financial calculations.

Only authorized users (admin) can manage system-level features such as monitoring user activities, managing application modules, and maintaining system data. Regular users can only access financial calculators and their own calculation history.

2.4 Assumptions and Dependencies

- Internet connectivity is required for communication with the cloud server for user authentication, data storage, and synchronization of financial calculation history.
- The application depends on Flutter framework for building a cross-platform mobile application that can run on both Android and iOS devices.
- The system relies on cloud services (such as Firebase) for backend operations including user authentication, database management, and secure data storage.
- Each user must have a unique account ID to access the application features and store their financial calculation records.
- Users are expected to enter accurate financial parameters such as loan amount, interest rate, and tenure to obtain correct calculation results.
- The system assumes that the user device meets the minimum hardware and software requirements required to run the application smoothly.

CHAPTER 3

SYSTEM DESIGN

3.1 UML Diagram

3.1.1 Use-case Diagram

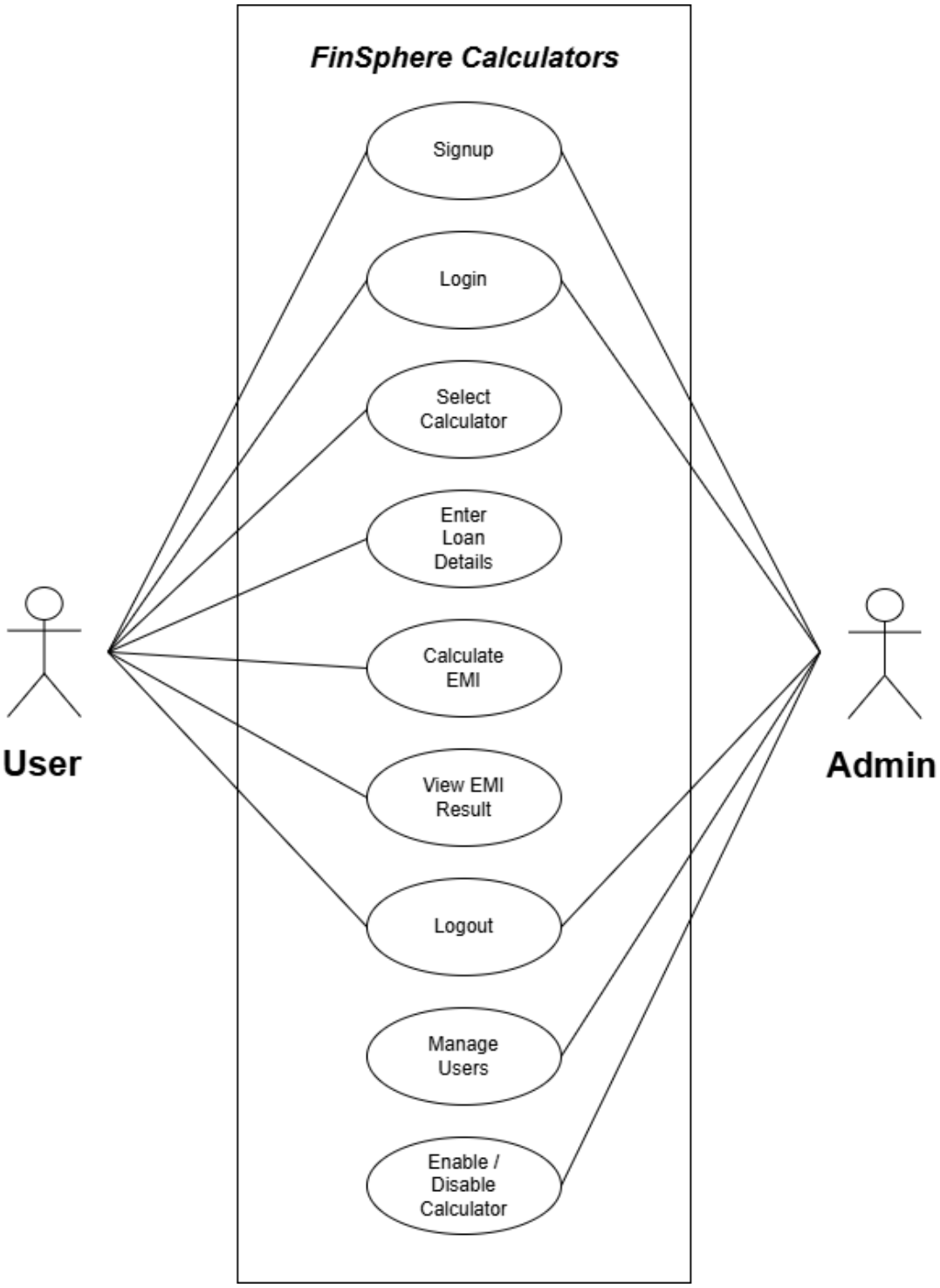


Figure 3.1.1.1 Use case diagram

3.1.2 Use-case Scenarios

Login

Field	Description
Abbreviated Title	Login
Actors	Admin, Registered User
Description	To access the application, the system validates the user's credentials. After successful authentication, users can perform financial calculations and access their account features.
Pre-Conditions	User must have the application installed and must be registered in the system.
Task Sequence	• System displays the login screen • User enters username/email and password. • System verifies the entered credentials. • System grants access to the user dashboard.
Post Conditions	System transfers control to the main dashboard where the user can access financial calculators and other features.
Exception	If the credentials are incorrect, the system displays an **"Invalid username or password"** message and returns to the login screen.

Perform Financial Calculation

Field	Description
Abbreviated Title	Financial Calculation
Actors	Registered User
Description	This use case allows users to calculate financial values such as loan EMI, FD maturity value, and RD returns using the calculator modules.
Pre-Conditions	User must be logged into the application.
Task Sequence	• System displays the list of available financial calculators. • User selects a calculator (e.g., EMI Calculator). • User enters required financial parameters such as principal amount, interest rate, and tenure • System processes the data and performs the calculation.
Post Conditions	The system displays calculation results such as EMI amount, total interest, and total repayment.
Exception	If invalid data is entered, the system displays an error message and asks the user to re-enter valid information.

Compare Loan Options

Field	Description
Abbreviated Title	Loan Comparison
Actors	Registered User
Description	This use case allows users to compare multiple loan options based on interest rate, tenure, and EMI amount.
Pre-Conditions	User must be logged in.
Task Sequence	• System displays loan comparison screen. • User enters details of multiple loan options. • System calculates EMI and interest for each option. • System compares results and displays comparison chart or table.
Post Conditions	User can analyze and choose the most suitable loan option.
Exception	If required parameters are missing, the system prompts the user to complete all fields.

Manage User Data

Field	Description
Abbreviated Title	Manage User Data
Actors	Admin
Description	Admin manages user accounts and system data within the application.
Pre-Conditions	• Admin must log in with administrator credentials.
Task Sequence	• System displays list of registered users. • Admin can view, update, or remove user records. • Admin can monitor system usage and activity logs.
Post Conditions	System updates the database with the changes made by the admin.
Exception	If an invalid operation occurs, the system displays an error message and cancels the action.

3.1.3 Activity Diagram

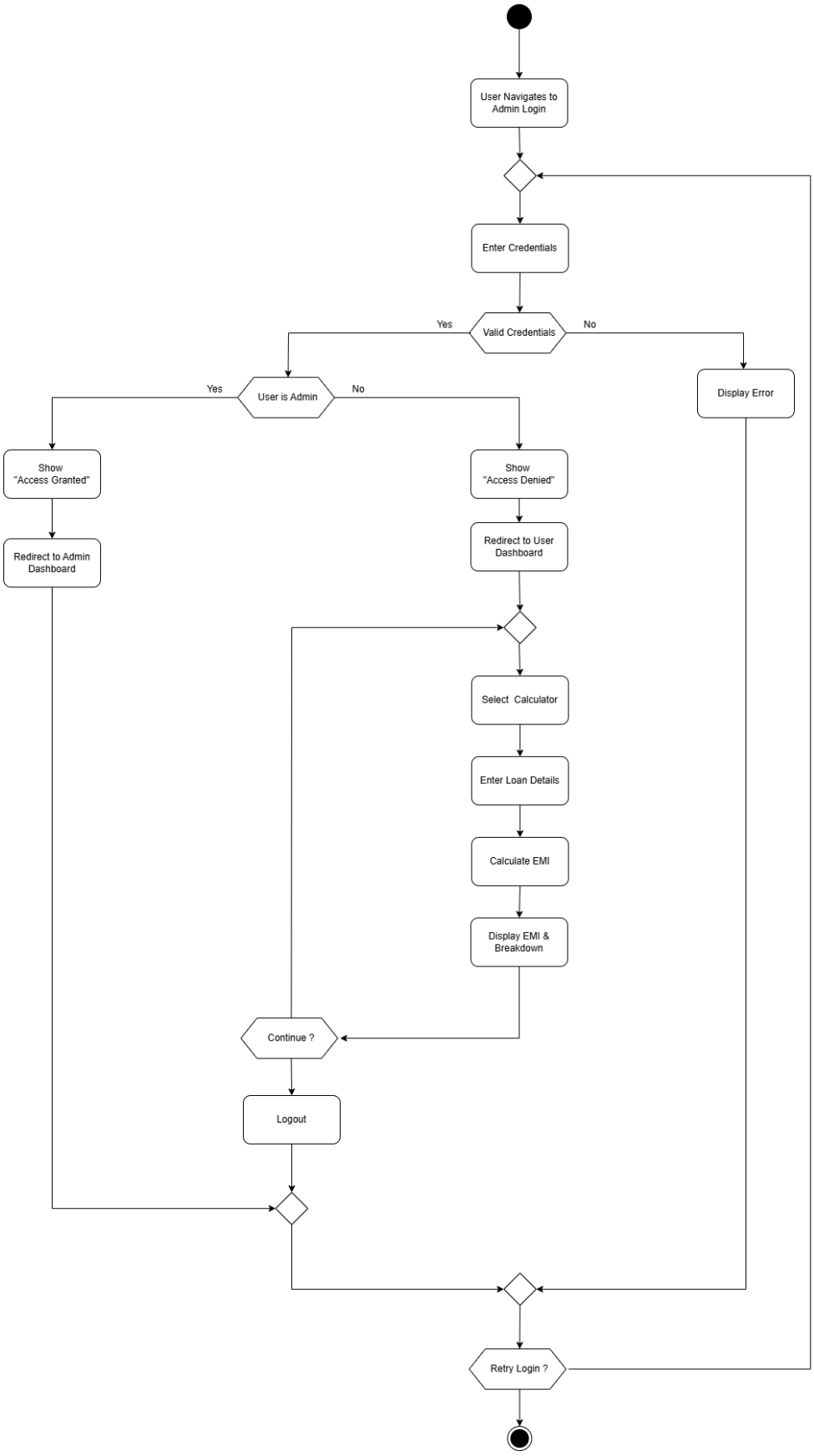


Figure 3.1.3.1 Activity diagram for Book Issue

3.1.4 Sequence Diagram

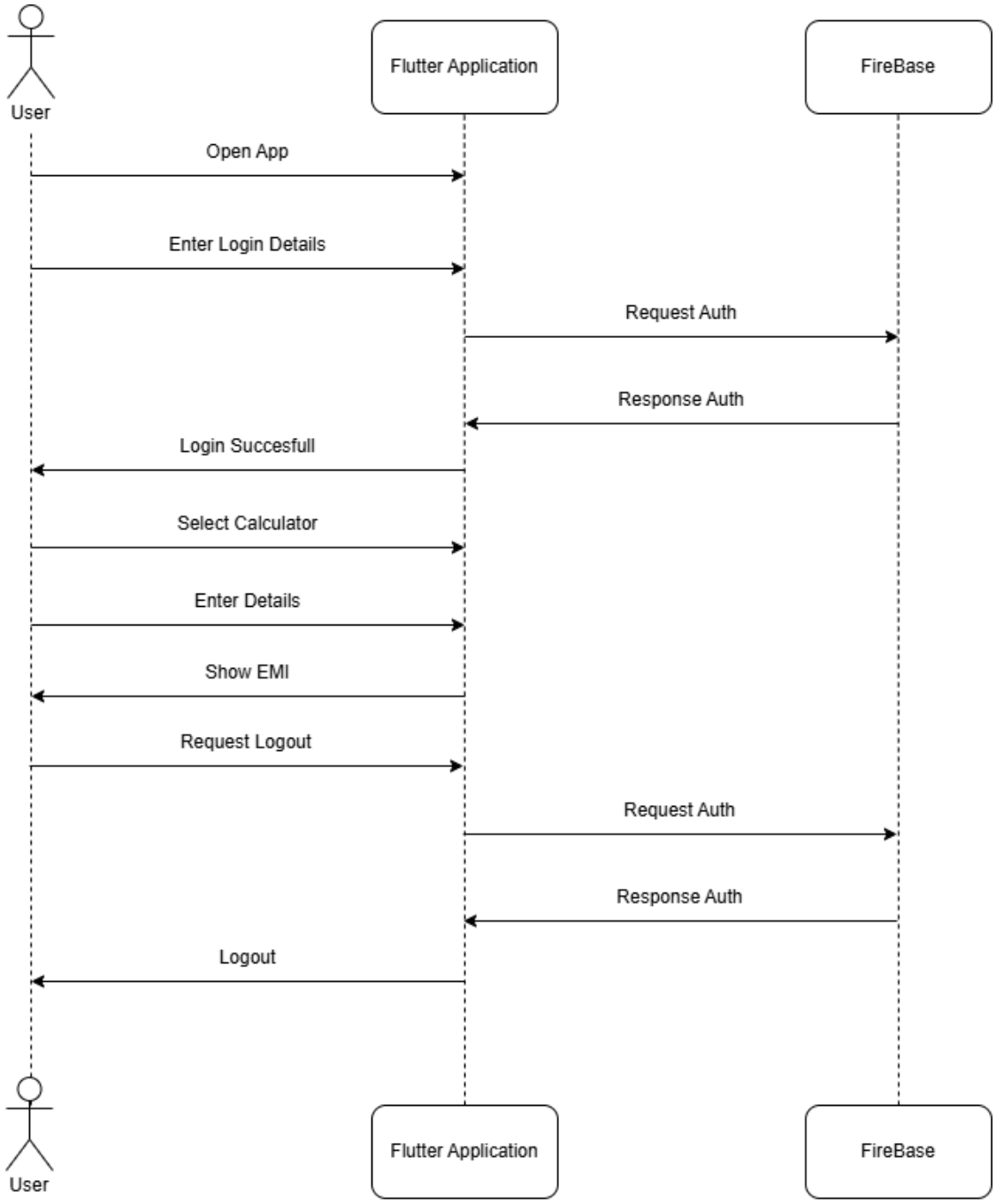


Figure 3.1.4.1 Sequence diagram for Book Issue

3.1.5 State Diagram

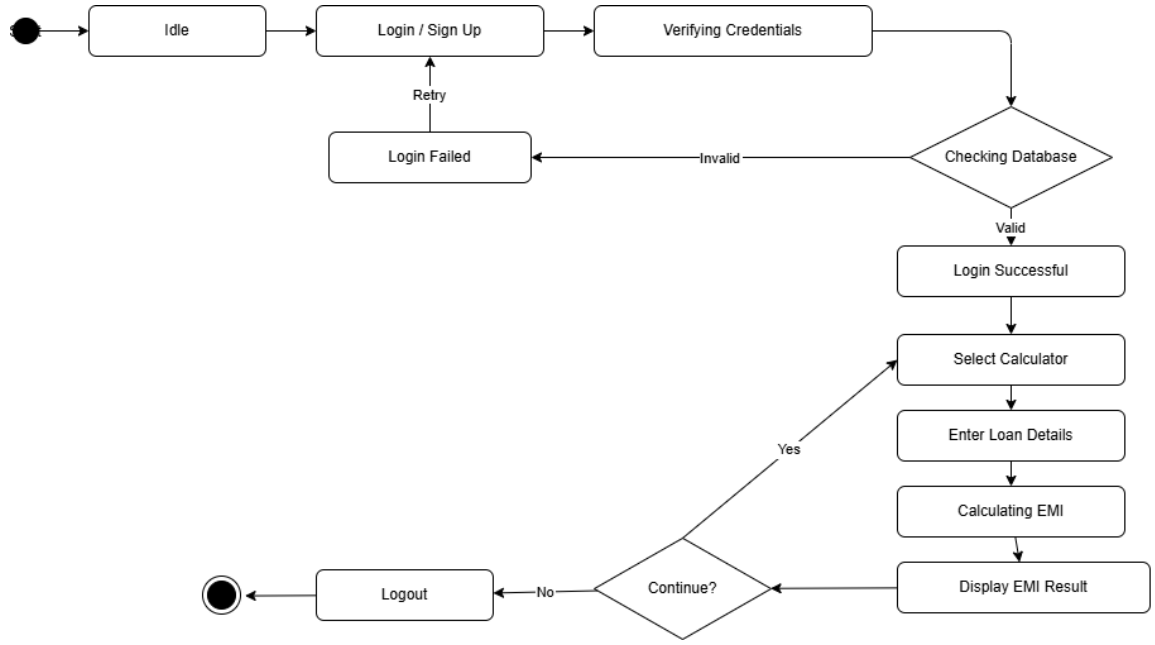


Figure 3.1.5.1 State diagram of Book

3.2 Database Design

3.2.1 Database Schema

Table 1: Admin

Column	Data Type	Null	Keys & constrains	Default Value & Description
UID	string	NN	PK (Auto Increment)	Unique Id Generated by Firebase
name	string	NN		Full Name of Admin
email	string	NN		Email Address of Admin
password	sring	NN		Password
enroll	string	NN		Admin Role (Default = “Admin”)

Table 3.2.1 Table: Admin

Table 2: User

Column	Data Type	Null	Keys & constrains	Default Value & Description
UID	string	NN	PK (Auto Increment)	Unique Id Generated by Firebase
name	string	NN		Full Name of Admin
email	string	NN		Email Address of Admin
password	sring	NN		Password
enroll	string	NN		User Role (Default = “User”)

Table 3.2.2 Table: User

3.2.2 E-R Diagram



CHAPTER 4

IMPLEMENTATION

4.1 Module Design and Specification

4.1.1 Screen-1: Login Screen



Figure 4.1 1 Screen-1: Login Screen

Purpose: This screen allows registered users or administrators to log into the FinSphere Calc system. Users must enter valid credentials such as email and password to access the financial calculator features of the application.

Sr.	Screen Element	Input Type	O/M	Description
1	Login Role (User/Admin)	Toggle Button	M	Allows the user to select login type either **User** or **Admin**.
2	Email Address	Textbox	M	Email field should be editable and must accept a valid email format.
3	Password	Password	M	Password field should be editable and display characters as dots or stars for security.
4	Show/Hide Password	Icon Button	O	Allows user to toggle password visibility.
5	Login	Button	M	Login button validates credentials and grants access to the system dashboard.
6	Register Now	Link	O	Redirects new users to the registration page to create a new account.

Table 4.1 1 Screen element of Login Screen

Screen-2: Signup Screen

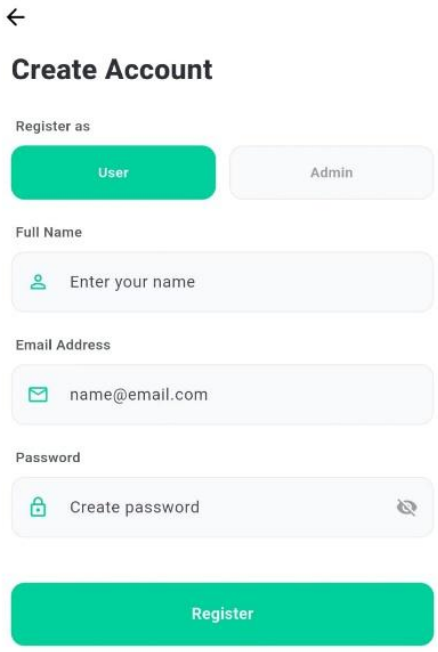


Figure 4.1.2 Screen-2: Signup Screen

Purpose: This screen allows new users to create an account in the FinSphere Calc application. Users must enter required personal information such as full name, email address, and password to register in the system. After successful registration, users can log in and access financial calculators and other application features.

Sr.	Screen Element	Input Type	O/M	Description
1	Register Role (User/Admin)	Toggle Button	M	Allows the user to select whether they are registering as a **User** or **Admin**.
2	Full Name	Textbox	M	Full name field should be editable and accept the user's name.
3	Email Address	Textbox	M	Email field should accept a valid email address format.
4	Password	Password	M	Password field should accept a secure password and display characters as dots or stars.
5	Show/Hide Password	Icon Button	O	Allows the user to toggle the visibility of the password.
6	Register	Button	M	Register button stores the entered data into the database and creates a new user account.
7	Back Arrow	*Icon Button*	O	Allows the user to navigate back to the previous screen (Login Screen).

Table 4.1.2 Screen element of Signup Screen

4.1.2 Screen-3: Dashboard

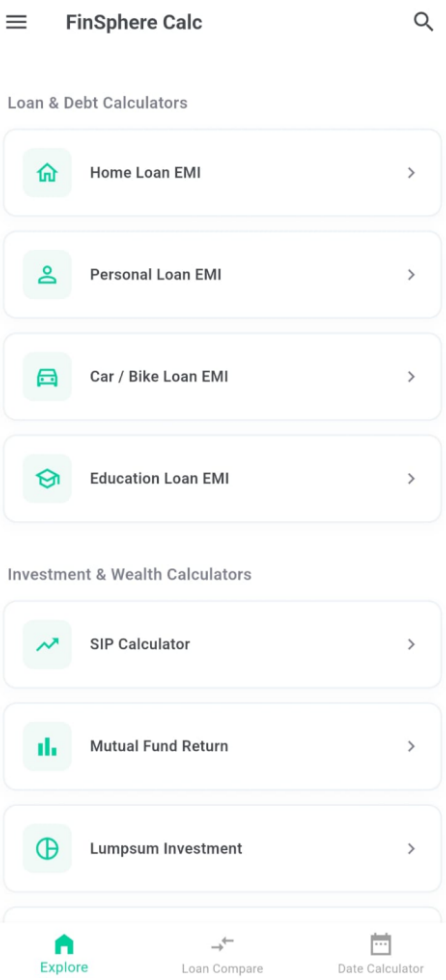


Figure 4.1.3 Screen-3: Dashboard

Purpose: This screen acts as the main dashboard of the FinSphere Calc application. It provides users with access to different financial calculators available in the system. Users can easily navigate through different categories such as Loan & Debt Calculators and Investment & Wealth Calculators to perform various financial calculations.

Sr.	Screen Element	Input Type	O/M	Description
1	Menu Icon	Icon Button	O	Opens the navigation drawer containing additional application options.
2	Search Icon	Icon Button	O	Allows users to search for specific calculators or features.
3	Loan & Debt Calculators Section	Label	O	Displays the category containing loan-related calculators.
4	Home Loan EMI	Navigation Card	O	Opens the Home Loan EMI Calculator screen.

5	Personal Loan EMI	Navigation Card	O	Opens the Personal Loan EMI Calculator screen.
6	Car/Bike Loan EMI	Navigation Card	O	Opens the Vehicle Loan EMI Calculator screen.
7	Education Loan EMI	Navigation Card	O	Opens the Education Loan EMI Calculator screen.
8	Investment & Wealth Calculators Section	Label	O	Displays the category containing investment-related calculators.
9	SIP Calculator	Navigation Card	O	Opens the Systematic Investment Plan (SIP) Calculator.
10	Mutual Fund Return	Navigation Card	O	Opens the Mutual Fund Return Calculator.
11	Lumpsum Investment	Navigation Card	O	Opens the Lumpsum Investment Calculator.
12	Bottom Navigation – Explore	Navigation Button	O	Opens the main calculator dashboard.
13	Bottom Navigation – Loan Compare	Navigation Button	O	Opens the Loan Comparison Tool.
14	Bottom Navigation – Date Calculator	Navigation Button	O	Opens the Date Calculation Tool.

Table 4.1.3 Screen element of Dashboard Screen

Screen-4: Calculator Screen

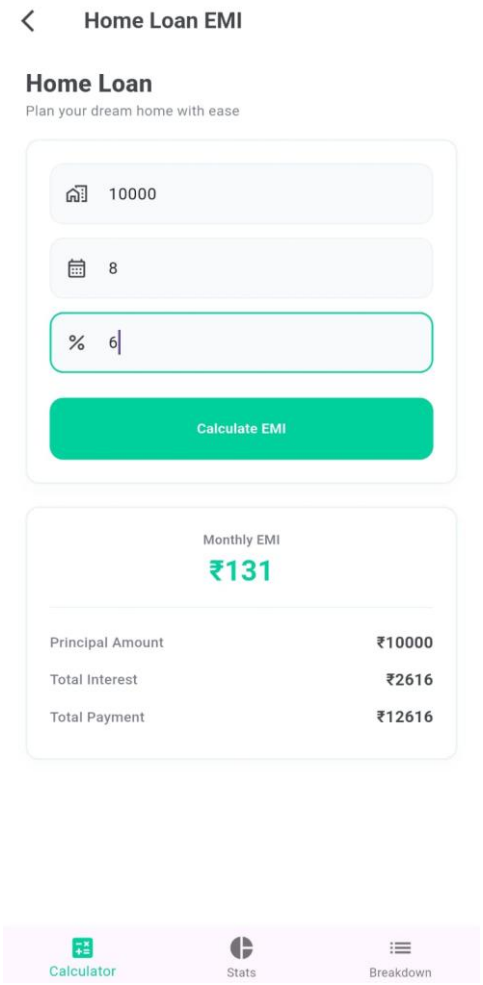


Figure 4.1.4 Screen-4: Calculator Screen

Purpose: This screen allows users to calculate the **Equated Monthly Installment (EMI)** for a home loan. Users can enter the **loan amount, loan tenure, and interest rate** to calculate the monthly EMI. The system then displays the **monthly EMI amount along with total interest and total payment details**, helping users understand their loan repayment structure.

Sr.	Screen Element	Input Type	O/M	Description
1	Back Button	Icon Button	O	Allows the user to return to the previous screen.
2	Loan Amount	Textbox	M	Field to enter the total loan amount required by the user.
3	Loan Tenure	Textbox	M	Field to enter the loan duration in years or months.
4	Interest Rate	Textbox	M	Field to enter the annual interest rate percentage.

5	Calculate EMI	Button	M	Button used to perform EMI calculation based on entered values.
6	Monthly EMI Result	Label	O	Displays the calculated monthly EMI amount.
7	Principal Amount	Label	O	Shows the principal loan amount entered by the user.
8	Total Interest	Label	O	Displays the total interest payable over the loan tenure.
9	Total Payment	Label	O	Displays the total amount payable (principal + interest).
10	Bottom Navigation – Calculator	Navigation Button	O	Opens the calculator section of the application.
11	Bottom Navigation – Stats	Navigation Button	O	Displays statistical charts or loan analysis.
12	Bottom Navigation – Breakdown	Navigation Button	O	Shows detailed payment breakdown information.

Table 4.1.4 Screen element of Calculator Screen

Screen 5 -: Pie Chart Screen

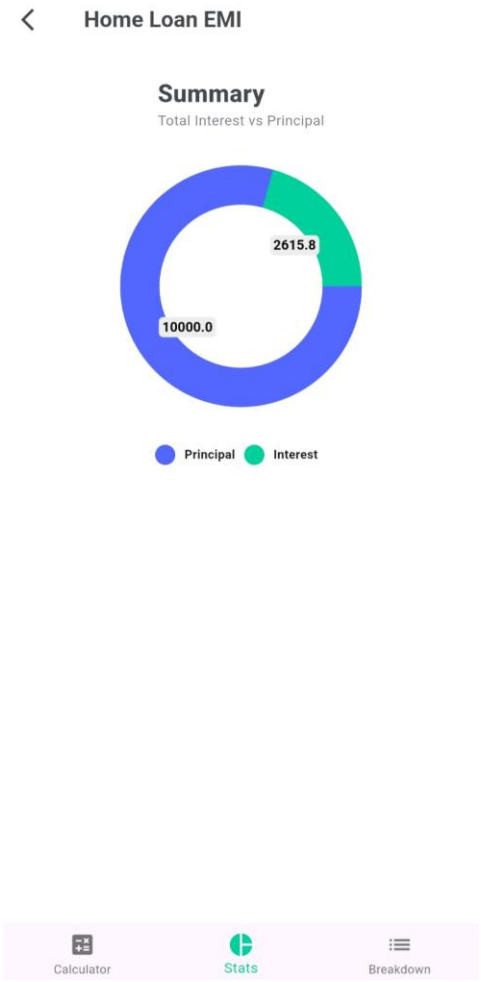


Figure 4.1.5 Screen-5: Pie Chart Screen

Sr.	Screen Element	Input Type	O/M*	Description
1	Back Arrow	Icon Button	O	Navigates the user back to the Home or Calculator input screen.
2	Screen Title	Label	M	Displays the specific calculator type (e.g., "Home Loan EMI").
3	Summary Chart	Donut Chart	M	Visually represents the ratio of Total Principal vs. Total Interest.
4	Data Labels	Text Overlay	M	Displays specific numerical values (Principal and Interest) directly on the chart segments.
5	Legend	Indicator	M	Color-coded labels (Principal: Blue, Interest: Green) to identify chart segments.
6	Bottom Nav Bar	Navigation Bar	M	Contains tabs for 'Calculator', 'Stats', and 'Breakdown' to switch views.

Table 4.1.5 Screen element of Pie Chart Screen

Screen 6 -: Date Difference Screen

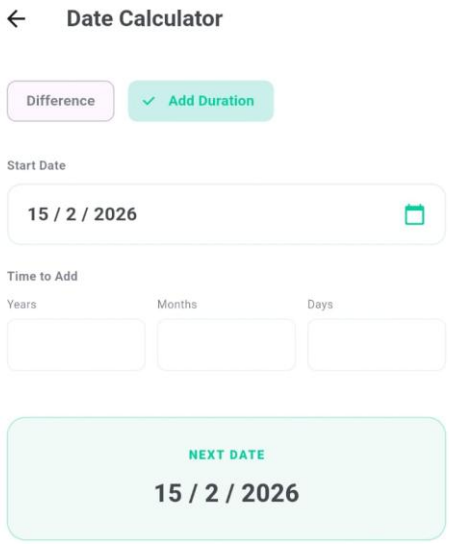


Figure 4.1.6 Screen-6: Date Calculator Screen

Sr.	Screen Element	Input Type	O/M*	Description
1	Back Arrow	Icon Button	O	Allows the user to navigate back to the main calculator menu or previous screen.
2	Screen Title	Label	M	Displays "Date Calculator" to identify the current tool being used.
3	Mode Selector	Toggle / Tab	M	Allows the user to switch between "Difference" (calculating days between dates) and "Add Duration".
4	Start Date	Date Picker	M	An input field that opens a calendar dialog to select the initial reference date.
5	Years Input	Textbox (Num)	O	Numerical input field to specify the number of years to add to the start date.
6	Months Input	Textbox (Num)	O	Numerical input field to specify the number of months to add to the start date.
7	Days Input	Textbox (Num)	O	Numerical input field to specify the number of days to add to the start date.
8	Result Card	Display Box	M	A highlighted container that shows the calculated "Next Date" based on the user's input.

Table 4.1.6 Screen element of Date Calculator

Screen 7 -: Loan Comparison Screen

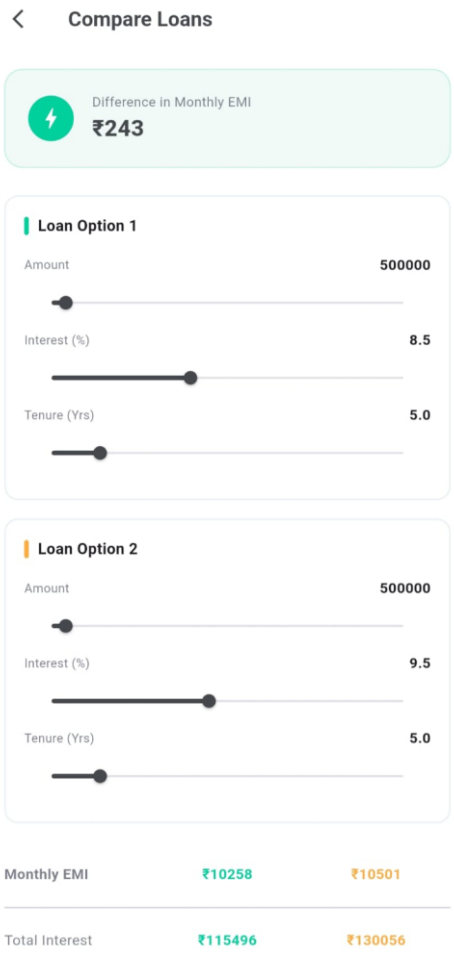


Figure 4.1.7 Screen-7: Loan Comparison Screen

Sr.	Screen Element	Input Type	O/M*	Description
1	Back Arrow	Icon Button	O	Navigates the user back to the previous screen or main dashboard.
2	Screen Title	Label	M	Displays "Compare Loans" to identify the current functionality.
3	EMI Difference Card	Display Card	M	Shows the calculated difference in Monthly EMI between Option 1 and Option 2.
4	Loan Option Cards	Container	M	Two distinct sections (Loan Option 1 & 2) grouped to hold separate sets of loan parameters.
5	Amount Sliders	Slider + Text	M	Dual-input controls to adjust the principal loan amount for both options.
6	Interest Sliders	Slider + Text	M	Dual-input controls to adjust the annual interest rate (%) for both options.

7	Tenure Sliders	Slider + Text	M	Dual-input controls to adjust the loan duration in years for both options.
8	Comparison Table	Data Grid	M	A bottom-aligned table comparing "Monthly EMI" and "Total Interest" for both loans side-by-side.

Table 4.1.7 Screen element of Loan Comparison Screen

4.2 Testing

4.2.1 Functional Testing

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC001	Open Calculator App	1. Launch the FinSphere Calc application	Application home screen is displayed successfully	Pass
TC002	Navigate to Home Loan EMI	1. Open app 2. Click "Home Loan EMI" option	Home Loan EMI calculator screen opens	Pass
TC003	Calculate Home Loan EMI	1. Enter Loan Amount 2. Enter Loan Tenure 3. Enter Interest Rate 4. Click "Calculate EMI"	System displays Monthly EMI, Total Interest, and Total Payment	Pass
TC004	Calculate Personal Loan EMI	1. Select Personal Loan EMI 2. Enter required values 3. Click Calculate	System calculates and displays correct EMI result	Pass
TC005	Calculate Car/Bike Loan EMI	1. Select Car/Bike Loan EMI 2. Enter loan details 3. Click Calculate	System displays calculated EMI result	Pass
TC006	Calculate SIP Investment	1. Open SIP Calculator 2. Enter monthly investment, rate and time period 3. Click Calculate	System displays total investment value and returns	Pass
TC007	Navigate Between Sections	1. Use bottom navigation (Explore / Loan Compare / Date Calculator)	System navigates to the selected section successfully	Pass

4.2.2 Non-Functional Testing

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC008	Performance Testing	Perform multiple EMI calculations repeatedly	System responds quickly without delay	Pass
TC009	Input Validation Testing	Enter invalid values (letters or empty fields)2. Click Calculate	System displays validation error or prevents calculation	Pass
TC010	Usability Testing	1. Test UI navigation 2. Check ease of entering data	Users can easily navigate and use calculator features	Pass
TC011	Compatibility Testing	Run the app on different Android devices	App works correctly on different screen sizes and devices	Pass

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

5.1 Conclusion

The FinSphere Calc application developed in this project successfully provides a digital platform for performing various financial calculations such as loan EMI calculations and investment estimations**.** The application simplifies complex financial computations and presents results instantly, allowing users to easily plan their loans and investments. By integrating multiple calculators such as Home Loan EMI, Personal Loan EMI, Car/Bike Loan EMI, Education Loan EMI, SIP Calculator, Mutual Fund Return Calculator, and Lumpsum Investment Calculator, the system offers a comprehensive financial calculation tool in a single application.

The system provides a user-friendly interface where users can enter basic financial inputs like loan amount, interest rate, and tenure to quickly calculate EMI and view detailed results including monthly EMI, principal amount, total interest, and total payment**.** This helps users make better financial decisions by understanding their repayment structure clearly.

Through this project, we have demonstrated how mobile technology can simplify financial planning and calculations**.** By replacing manual calculations with an automated and accurate system, the application improves efficiency, reduces calculation errors, and provides quick financial insights. The project reflects the importance of digital financial tools in modern life**,** making financial planning easier, faster, and more accessible for users.

5.2 Future Enhancement

While the current **FinSphere Calc application** provides essential financial calculation tools, several enhancements can be implemented in the future to improve its functionality, accuracy, and user experience.

1. **Advanced Financial Calculators**
   Additional financial calculators can be integrated into the application, such as **tax calculators, retirement planning calculators, savings goal calculators, and inflation calculators**. This will make the application a more comprehensive financial planning tool for users.

2. **R Graphical Data Visualization**
   Future versions of the system can include **interactive charts and graphs** to visually represent loan repayment schedules, investment growth, and interest distribution. Graphical visualization will help users better understand their financial data.

3. **Loan Comparison Feature**
   A **loan comparison module** can be added where users can compare different loan options from various banks based on interest rates, tenure, and EMI amounts. This will help users choose the most suitable loan option.

APPENDICES

Appendix A: Acronyms and Abbreviations

Acronym	Full Form
EMI	Equated Monthly Installment
UI	User Interface
UX	User Experience
API	Application Programming Interface
SIP	Systematic Investment Plan
ROI	Return on Investment
UML	Unified Modeling Language
Context	A high-level diagram representing the overall structure and interaction of the system
Use Case	Describes the interaction between users and the system to achieve a specific task

REFERENCES

[1] Flutter Learning Resources
https://docs.flutter.dev/learn/learning-resources

[2] Firebase – Riwan Ranawat (YouTube)
https://youtu.be/LFlE8yV7lJY?si=HdUVKQmzCNbP4X1c

[3] Chart Animation – Techie Blossom (YouTube)
https://youtu.be/8wXpkv0Mh38?si=Aqib9kJzVDlEGwaA